Assignment 3

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if you idea is (mostly) correct and you haven showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (10 marks) Let ε be the elliptic curve $y^2 = x^3 + x + 6$ over $\mathbb{F}_{11}$ and $P = (2, 7), Q = (5, 2)$. Compute $P + Q$.

Problem 2 (10 marks) Let ε be the elliptic curve $y^2 = x^3 + x + 4$ over $\mathbb{F}_5$. Show that $(3, 2)$ is a 3-torsion point.

Problem 3 (10 marks) A pairing $e : G_1 \times G_2 \to G_T$ is said to be *alternating* if $G_1 = G_2$ and $e(g, g) = 1$ for all $g \in G_1$. Prove that if e is alternating, then $e(g, h) = e(h, g)^{-1}$.

Problem 4 (10 marks) Assume G_1, G_2, G_T are cyclic. Prove that the following pairing inversion is not harder than DL problem in G_T.
Definition(Pairing inversion): Let $G_1 \times G_2 \to G_T$ is a pairing. Given $g_t \in G_T$ and $h \in G_2$, find $g \in G_1$ such that $e(g, h) = g_t$.

Problem 5 (20 marks) Let $E[N]$ denote the N-torsion subgroup of E, and let (P, Q) be a basis of $E[N]$, so that every point in $E[N]$ can be written uniquely as $aP + bQ$ for some $a, b \in \mathbb{Z}/N\mathbb{Z}$. Let $e : E[N] \times E[N] \to \langle \xi \rangle_N$ be an alternating pairing. Show that $e(aP + bQ, cP + dQ) = e(P, Q)^{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$ mod N, where $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

Problem 6 (20 marks) Suppose that $p > 3$ is an odd prime, and $a, b \in \mathbb{Z}_p$ and the equation $x^3 + ax + b \equiv 0 \mod p$ has three distinct roots in $\mathbb{Z}_p$. Show that the corresponding elliptic curve group $(\varepsilon, +)$ is not cyclic.(Hint: show that the points of order two generate a subgroup of $(\varepsilon, +)$ that is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.)

Problem 7 (20 marks) Consider an elliptic curve ε described by the formula $y^2 \equiv x^3 + ax + b \mod p$, where $4a^3 + 27b^2 \neq 0 \mod p$ and $p > 3$ is prime.

- It is clear that a point $P = (x_1, y_1) \in \varepsilon$ has order 3 if and only if $2P = -P$. Use this fact to prove that, if $P = (x_1, y_1) \in \varepsilon$ has order 3, then
$$3x_1^4 + 6ax_1^2 + 12x_1 b - a^2 \equiv 0 \mod p.$$
- Conclude from the above equation that there are at most 8 points of order 3 on the elliptic curve ε.